FAQs

1. What is a tree?

Ans: A tree is a non-linear data structure used for hierarchical data management.

2. What is a binary tree?

Ans: A binary tree is a tree in which each node has at most two children, called the left child and right child.

3. What are the types of binary trees?

Ans: Complete binary tree, full binary tree, binary search tree, skewed binary tree and threaded binary tree.

4. What are tree traversal techniques?

Ans: Tree traversal techniques are of two types: recursive and non-recursive.

5. What are recursive tree traversal techniques?

Ans: The three recursive tree traversal techniques are:

1. Inorder
2. Preorder
3. Postorder

6. What are AVL trees?

Ans: AVL trees are self-balancing binary search trees in which the height difference between the left and right subtrees is at most 1.

7. What are the basic operations that can be performed on a tree?

Ans: The basic operations are creation, insertion, searching, deletion and traversal.

8. Explain the terms degree and height of a tree.

Ans:

- Degree: The number of children of a node.

- Height: The maximum number of edges from the root to a leaf node.

9. Draw the representation of a tree using a linked list and an array.

Ans:

- Linked list representation: Each node contains data and pointers to its left and right children.
- Array representation: Tree nodes are stored in an array, with child positions calculated using indices.